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A lightweight high-resolution feature aggregation design for animal pose estimation

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ABSTRACT

Animal pose estimation is increasingly deployed in behavior analysis and monitoring, where real-time inference on resource-constrained devices is required. Compared to human pose estimation, animal pose estimation involves diverse species with distinct anatomical structures and significant pose variations across species, requiring both cross-species generalization and spatial detail preservation. Existing lightweight methods often struggle to balance computational efficiency and high-resolution representation, mainly due to inefficient multi-scale feature aggregation. To address this issue, we propose a single-pass high-resolution feature calibration strategy, which aligns and aggregates multi-scale features into a unified high-resolution representation in an efficient manner. This design is instantiated within a lightweight framework (HRCPose) for accurate keypoint localization under constrained computation. Experiments on AP-10K and a real-world macaque cage-surveillance dataset demonstrate consistent improvements in localization accuracy under comparable computational budgets. Additional COCO results show the method generalizes beyond animal scenarios.

1. Introduction

In practical animal monitoring systems, pose estimation models are often deployed on edge devices with limited computational resources, where maintaining localization accuracy under real-time constraints remains a challenging problem. Essentially, the task involves detecting anatomical landmarks in images and reconstructing their spatial configuration. Unlike human pose estimation, animal pose estimation must cope with substantial inter-species appearance differences and highly variable body structures, making robust feature representation and precise keypoint localization considerably more challenging. Rao et al. [1], Ye et al. [2] While recent deep learning methods have achieved strong accuracy, most high-performing approaches rely on heavyweight architectures. Representative top-down models, such as HRNet, SimpleBaseline, and ViTPose, employ high-resolution parallel networks, deep residual backbones, or vision transformers to preserve detailed spatial representations [3–5]. These designs preserve detailed spatial information and achieve high localization accuracy, but incur substantial computational overhead, limiting their applicability in real-time scenarios.

To reduce computational cost, lightweight pose estimators have been proposed. Coordinate-classification-based methods such as RTMPose [6] achieve competitive accuracy with high efficiency. However, under strict computational budgets, they often struggle to preserve fine-grained spatial details, mainly due to limitations in feature aggregation. In contrast, lightweight high-resolution networks such as LiteHRNet [7,8] maintain stronger spatial representations through parallel multi-resolution branches, but their parallel design and repeated feature fusion introduce additional overhead and limit inference efficiency. A common limitation of current lightweight architectures lies in how multi-scale information is integrated. Feature interactions are often either delayed until late stages or achieved through repeated fusion operations, which may weaken cross-scale information flow and introduce unnecessary computational overhead.

To address the loss of fine-grained spatial details in animal pose estimation, a single-pass high-resolution feature calibration strategy is introduced. As shown in Fig. 1, a clear trade-off exists. The proposed design aligns and aggregates multi-scale features into a unified high-resolution representation in a single step, reducing information loss while maintaining computational efficiency. Unlike conventional multi-stage fusion strategies, the proposed approach avoids repeated feature transformations, which helps preserve spatial consistency under lightweight constraints. This design is instantiated within a lightweight

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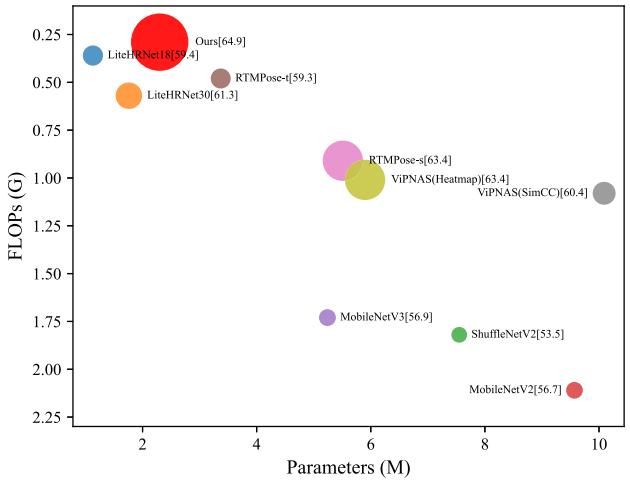


Fig. 1. Accuracy–efficiency comparison of different models on the AP-10K dataset. The x-axis denotes the number of parameters, the y-axis denotes FLOPs, and the bubble size represents AP. Larger bubble indicates higher AP.



Fig. 2. Qualitative pose estimation results of HRCPose on AP-10K across diverse animal species and poses.

framework, termed HRCPose, to support efficient feature extraction and coordinate prediction. Qualitative results in Fig. 2 show consistent and complete predictions across diverse species.

To evaluate deployment-oriented scenarios, we construct CM-9k, a macaque pose dataset collected from cage-surveillance environments. The dataset reflects realistic conditions such as complex illumination and pose variations, where preserving fine-grained spatial details is critical. Together with AP-10K, it enables evaluation under both cross-species and in-domain settings, while experiments on COCO further verify that the proposed design generalizes to human pose estimation scenarios. The main contributions of this work are summarized as follows:

- We propose a lightweight high-resolution pose estimation framework with a single-pass feature aggregation design, which efficiently preserves fine-grained spatial details under constrained computation.
- A macaque pose dataset, CM-9k, is constructed from real-world cage-surveillance scenarios, covering complex illumination and pose variations.
- Extensive experiments on AP-10K, CM-9k, and COCO demonstrate consistent improvements in localization accuracy under comparable computational budgets.

2. Related work

2.1. Animal pose estimation

Animal pose estimation has been widely studied for behavior analysis, health monitoring, and biological research [1,9–12]. With the development of deep learning, top-down frameworks such as SimpleBaseline, HRNet, and ViTPose achieve strong performance by leveraging high-resolution representations and heatmap-based localization [3–5]. However, these methods rely on heavyweight backbones and dense prediction heads, resulting in high computational cost and limited applicability in real-time scenarios.

To improve efficiency, lightweight designs based on MobileNet, ShuffleNet, and LiteHRNet have been proposed [7,13–17]. While reducing model complexity, these approaches often suffer from degraded localization accuracy. LiteHRNet attempts to maintain high-resolution representations, but still exhibits limitations in accuracy and inference efficiency. Coordinate classification methods such as SimCC and RTMPose [6,18] further improve the accuracy–efficiency balance. Nevertheless, under strict computational budgets, existing methods still struggle to preserve fine-grained spatial details while maintaining efficient inference.

2.2. Animal pose datasets

Recent datasets, including AP-10K, OpenMonkeyChallenge, AnimalPose, and APt-36k [19–22], have significantly advanced animal pose estimation by providing diverse species coverage and standardized evaluation protocols. These benchmarks mainly focus on cross-species generalization and improve comparability across methods.

However, existing datasets provide limited coverage of macaque-centered data in indoor cage-surveillance environments, where data are typically captured from high-resolution surveillance sources and involve complex illumination and pose variations. Such conditions pose additional challenges for lightweight models due to the loss of fine-grained spatial details. To better evaluate these deployment-oriented scenarios, we construct the Caged Macaca fascicularis 9k (CM-9k) dataset, which complements existing benchmarks by providing a realistic testbed for high-resolution monitoring settings.

3. Datasets

Existing animal pose benchmarks emphasize cross-species diversity, but provide limited coverage of high-resolution cage-surveillance scenarios, which are essential for evaluating deployment-oriented monitoring systems. In particular, macaque-centered data captured in real surveillance environments remains scarce. To address this gap, we construct CM-9k, a macaque pose dataset collected from high-resolution surveillance videos. The data is captured by fixed cameras at a resolution of 3840×2160, and processed into standardized instance-level inputs for training and evaluation. The dataset contains 9270 annotated images. The dataset is split into training and validation sets with a ratio of 4:1.

CM-9k reflects realistic monitoring conditions, including complex illumination changes, occlusions, and diverse poses. These factors introduce significant challenges for lightweight pose estimation models, as the loss of fine-grained spatial details can lead to degraded localization accuracy.

We annotate 17 landmarks following the OpenMonkeyChallenge protocol [19], enabling consistent evaluation. As shown in Fig. 3, CM-9k provides a realistic testbed for evaluating pose estimation methods in high-resolution surveillance scenarios.

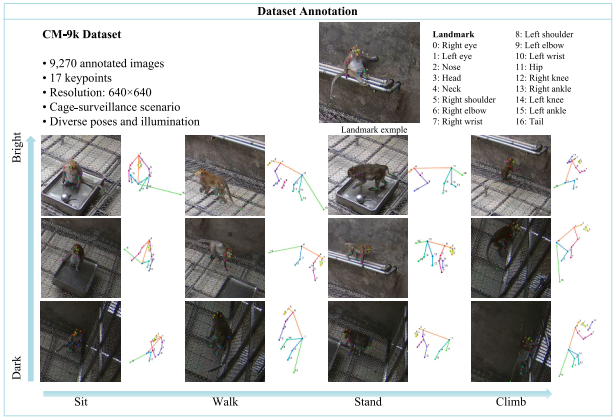


Fig. 3. Overview of our cynomolgus macaque pose dataset. The figure summarizes the construction pipeline from high-resolution surveillance frames to standardized instance crops, showcases representative samples under time-varying illumination and diverse behaviors, and illustrates the 17-landmark definition following the OpenMonkeyChallenge paradigm [19].

4. Method

In lightweight pose estimation, effective multi-scale feature aggregation is critical for preserving spatial precision under constrained computation. Following the lightweight coordinate-classification paradigm (e.g., RTMPose), we instantiate our design within a lightweight framework. Existing methods often rely on either delayed fusion or repeated multi-stage fusion, which limits efficiency or introduces redundancy.

Feature Pyramid Networks (FPN) perform fusion only after feature extraction, resulting in limited cross-scale interaction. Methods such as PAFPN improve information exchange through iterative top-down and bottom-up pathways, but incur repeated transformations and increased latency. More recent approaches, such as CEDNet [23], introduce earlier feature interaction through cascade structures, yet increase architectural complexity.

These observations suggest that efficient cross-scale interaction should be achieved without repeated feature transformations. We therefore adopt a single-pass high-resolution aggregation principle, where multi-scale features are projected into a unified high-resolution space and interacted in a single step, as illustrated in Fig. 4. Based on this principle, we design the Scale Calibration Adapter (SCA), and term the resulting framework HRCPose, where HRC denotes high-resolution calibration.

4.1. Scale calibration adapter

A key challenge in lightweight pose estimation is to consolidate multi-scale features without degrading high-resolution representations. Existing fusion architectures typically rely on iterative top-down and bottom-up propagation, which introduces additional computational overhead and may weaken spatial consistency.

We argue that effective cross-scale interaction does not necessarily require multiple propagation stages. If multi-scale features are first aligned into a common high-resolution space, feature interaction can be completed through a single aggregation operation. Based on this observation, we propose the Scale Calibration Adapter (SCA), which performs single-pass high-resolution aggregation for efficient cross-scale interaction without repeated transformations.

Specifically, given the backbone feature pyramid $\{P_3, P_4, P_5\}$, SCA aligns all features to the resolution of P_3 and projects them to a shared channel width $c = 128$:

$$R_3 = \mathrm{Conv}_{1\times1}(P_3) \in \mathbb{R}^{c\times H\times W} \tag{1}$$

$$R_4 = \mathrm{Up}\big(\mathrm{Conv}_{1\times1}(P_4)\big) \in \mathbb{R}^{c\times H\times W} \tag{2}$$

$$R_5 = \mathrm{Up}\big(\mathrm{Conv}_{1\times1}(P_5)\big) \in \mathbb{R}^{c\times H\times W} \tag{3}$$

To reduce inter-scale mismatch, we apply a shared lightweight calibration module composed of depthwise convolution and efficient channel attention (ECA) [24]. This shared design aligns feature distributions across scales while preserving efficiency. For a feature map F:

$$\mathrm{DWECA}(F) = F \odot \sigma\Big(\mathrm{Conv1D}\big(\mathrm{GAP}\big(\mathrm{DWConv}_{3\times3}(F)\big)\big)\Big) \tag{4}$$

After calibration, the aligned features are aggregated by element-wise summation in the unified high-resolution space, followed by normalization and activation:

$$Y = \mathrm{SiLU}\Big(\mathrm{BN}\big(\mathrm{DWECA}(R_3) + \mathrm{DWECA}(R_4) + \mathrm{DWECA}(R_5)\big)\Big) + \mathrm{DWECA}(R_3) \tag{5}$$

This single-pass aggregation preserves high-resolution spatial details while avoiding channel expansion and repeated fusion overhead. By performing feature interaction directly in the high-resolution space, it maintains spatial consistency and reduces information distortion, making it well-suited for efficient edge inference.

4.2. CSPFaster backbone

To support the proposed single-pass high-resolution aggregation, we adopt a lightweight CSPFaster backbone derived from YOLOv11n [29]. The backbone produces a compact feature pyramid $\{P_3, P_4, P_5\}$ with output strides of 8, 16, and 32, respectively. The corresponding channel dimensions are 128, 128, and 256. Accordingly, the spatial resolutions of P_3, P_4, and P_5 are $\frac{1}{8}$, $\frac{1}{16}$, and $\frac{1}{32}$ of the input image. These multi-scale features are subsequently fed into SCA for unified high-resolution aggregation. To enhance contextual modeling at the deepest stage, we replace the original SPPF block with a lightweight Depthwise Dilated Convolution Pyramid (DCP). DCP employs three parallel depthwise 3×3 convolutions with dilation rates $\{1, 2, 3\}$ to capture multi-scale contextual information. The branch outputs are aggregated and transformed as

$$Y = X + \mathrm{PWConv}_{1\times1}\Big(\mathrm{SiLU}\big(\mathrm{BN}(\mathrm{DWConv}^1_{3\times3}(X) + \mathrm{DWConv}^2_{3\times3}(X) + \mathrm{DWConv}^3_{3\times3}(X))\big)\Big) \tag{6}$$

where X and Y denote the input and output features, respectively. The residual connection preserves the original representation while the dilated depthwise branches provide efficient multi-scale context aggregation.

4.3. Structure-aware projection module

To complement the proposed high-resolution aggregation, we introduce a Structure-Aware Projection (SAP) module for efficient feature-to-coordinate mapping. SAP replaces the computationally expensive 7×7 convolution used in the RTMCC head [6] with a lightweight projection design.

Specifically, SAP first applies axial depthwise convolutions to capture directional context:

$$X^{\mathrm{ax}} = \mathrm{SiLU}\Big(\mathrm{BN}\big(\mathrm{DWConv}_{1\times7}\big(\mathrm{SiLU}\big(\mathrm{BN}\big(\mathrm{DWConv}_{7\times1}(X)\big)\big)\big)\big)\Big) \tag{7}$$

$$\hat{X} = X + \mathrm{SiLU}\Big(\mathrm{BN}\big(\mathrm{DWConv}_{3\times3}(X^{\mathrm{ax}})\big)\Big) \tag{8}$$

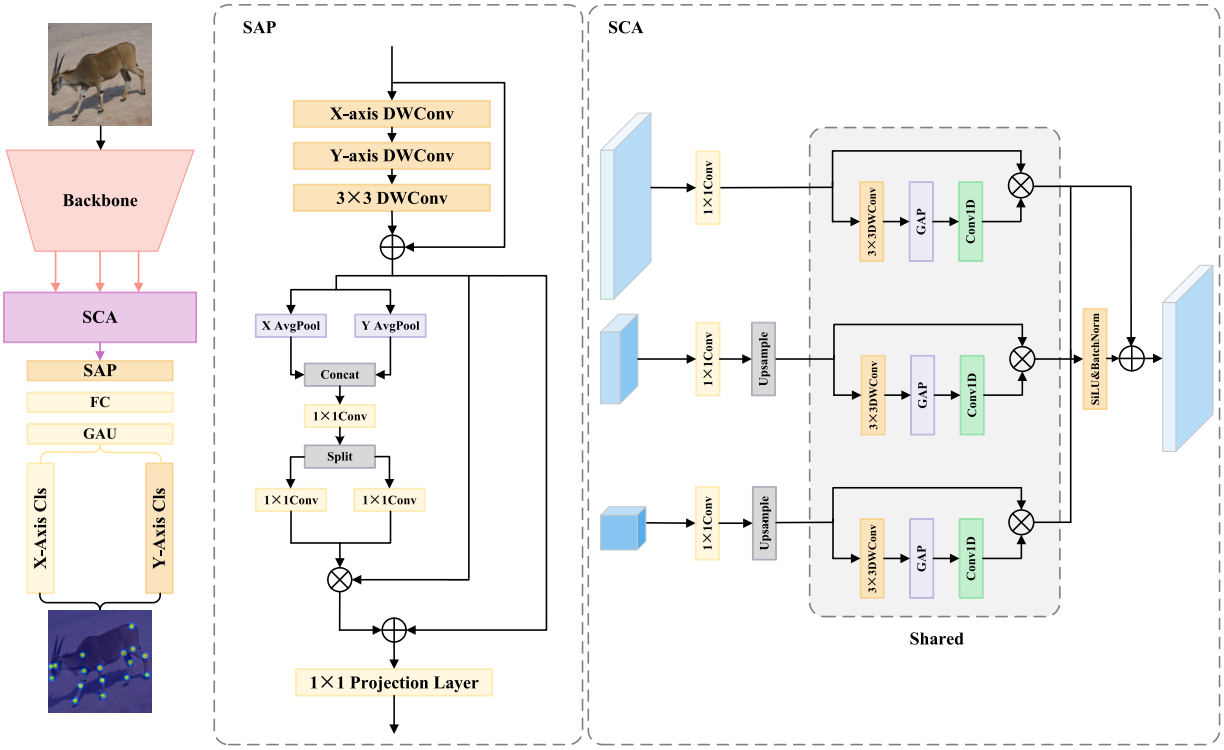


Fig. 4. Overview of HRCPose. The backbone extracts multi-scale features, SCA aggregates them into a high-resolution representation, and SAP with the head predicts keypoint coordinates.

We then apply coordinate attention [25] to enhance spatial awareness:

$$\tilde{X} = \text{CoordAttnRes}(\hat{X}) = \hat{X} + \hat{X} \odot A(\hat{X}) \tag{9}$$

Finally, a 1×1 convolution is applied to project the features into joint maps. This lightweight projection provides efficient coordinate prediction with low computational cost.

5. Experiments

Experiments are conducted on an NVIDIA GeForce RTX 2080 Ti GPU. For deployment evaluation, inference speed is measured on a Xiaomi 10 smartphone with a Qualcomm Snapdragon 865 processor. Performance is evaluated using average precision (AP) and average recall (AR), as well as MPJPE and PCK.

Implementation details. The proposed framework is implemented in PyTorch and trained using the AdamW optimizer. Detailed implementation settings and training hyper-parameters are provided in Table 1.

Table 1
Key implementation settings and training hyper-parameters.

Hyper-parameter	Value
Batch size	256
Training epochs	840
Optimizer	AdamW
Initial learning rate	4×10^{-3}
Weight decay	0.05
Warm-up iterations	2000
LR scheduler	Cosine Annealing
Loss function	KLDiscretLoss
SimCC split ratio	2.0
EMA momentum	2×10^{-4}
Data augmentation	Flip, rotation, scaling, HSV jitter
Inference strategy	Flip test
Workers	16

5.1. Comparisons with existing methods

This section evaluates HRCPose on AP-10K under the standard top-down protocol, with additional validation on COCO to assess generalization (Tables 2, 3).

HRCPose achieves 64.9 AP with only 0.29 GFLOPs on AP-10K, demonstrating a superior accuracy-efficiency trade-off. Compared with RTMPose-t, it improves AP by 5.6% while reducing FLOPs by 39.6%. The gains are consistent across localization metrics, indicating more accurate and stable predictions. On COCO, HRCPose maintains competitive performance under a similar computational budget, confirming good generalization.

These gains are mainly attributed to the proposed single-pass high-resolution aggregation design. By consolidating multi-scale features into a unified high-resolution representation, the SCA module effectively preserves fine-grained spatial details while avoiding redundant transformations. This enables accurate localization under constrained computational budgets. Fig. 5 presents qualitative comparisons with representative lightweight models. In challenging multi-animal scenes with occlusion or strong interactions, existing methods often miss keypoints due to the loss of fine spatial details. In contrast, HRCPose produces more complete and consistent predictions, demonstrating stronger robustness in preserving structural information. Even in simpler cases, it maintains better structural continuity, particularly for thin articulated parts such as limbs.

HRCPose also demonstrates strong runtime efficiency across all evaluated platforms (Table 4). It achieves the highest throughput on both GPU and CPU, and the fastest inference speed on mobile devices, indicating its suitability for real-time deployment.

5.2. Ablation and architecture analysis

We analyze the contribution of each component on AP-10K, as shown in Table 5. Introducing SCA brings the most significant improvement,

Table 2
Comprehensive comparison with lightweight pose estimation models on the AP-10K test set with input size of 256 × 256.

Model	Param ↓	FLOPs ↓	AP ↑	AR ↑	MPJPE ↓	PCK@0.1 ↑	PCK@0.2 ↑	PCK@0.5 ↑	Mean PCK ↑
LiteHRNet18 [7]	**1.13M**	0.36G	59.4	64.4	0.233	0.715	0.842	0.941	0.833
LiteHRNet30	1.76M	0.57G	61.3	66.4	**0.212**	**0.724**	0.852	0.945	0.840
ShuffleNetV2 [13]	7.55M	1.82G	53.5	59.1	0.255	0.674	0.817	0.931	0.807
MobileNetV2 [14]	9.57M	2.11G	56.7	61.8	0.235	0.691	0.833	0.941	0.822
MobileNetV3	5.24M	1.73G	56.9	61.5	0.244	0.686	0.829	0.937	0.817
RTMPose-t	3.37M	0.48G	59.3	63.0	0.230	0.687	0.840	0.943	0.823
RTMPose-s	5.51M	0.91G	63.4	66.7	0.223	0.712	0.855	0.948	0.838
ViPNAS(SimCC) [26]	10.09M	1.08G	60.4	64.4	0.226	0.692	0.844	0.945	0.827
ViPNAS(Heatmap)	5.90M	1.01G	63.4	67.5	0.222	0.722	0.857	0.948	0.842
HRCPose (Ours)	2.30M	**0.29G**	**64.9**	**68.2**	0.220	**0.724**	**0.858**	**0.950**	**0.844**

Table 3
Generalization comparison on the COCO val set with representative lightweight pose estimators with input size of 256 × 192.

Model	Params ↓	FLOPs ↓	AP ↑	AR ↑
LiteHRNet18	**1.13M**	**0.21G**	64.5	70.1
LiteHRNet30	1.76M	0.32G	67.1	73.0
MobileNetV3	5.24M	1.30G	66.3	69.8
ShuffleNetV2	7.55M	1.37G	60.0	63.9
RTMPose-t	3.34M	0.36G	66.1	69.2
RTMPose-s	5.47M	0.68G	**70.2**	**73.2**
ViPNAS(Heatmap)	5.90M	0.76G	69.5	72.7
HRCPose (Ours)	2.21M	0.22G	68.0	71.9

Table 4
Comparing the speed and model size of models under an input resolution of 384 × 288.

Model	FPS (GPU) ↑	FPS (CPU) ↑	FPS (Mobile Device) ↑	ModelSize ↓
LiteHRNet18	250.32	24.65	11.08	**4.88MB**
ViPNAS(Heatmap)	189.91	12.73	7.61	22.46MB
MobileNetV2	634.11	48.70	14.38	36.46MB
ShuffleNetV2	766.29	39.35	17.43	28.80MB
RTMPose-t	1033.98	130.43	33.92	13.23MB
RTMPose-m	451.37	53.64	11.84	52.31MB
HRCPose(Ours)	**1054.54**	**147.91**	**42.02**	9.80MB

Table 5
Ablation study on AP-10K.

SAP	SCA	DCP	AP(%) ↑	AR(%) ↑	FLOPs ↓	Params ↓
			58.8	62.4	0.273G	2.28M
✓			59.8	63.2	0.260G	2.10M
	✓		63.4	66.8	0.394G	2.40M
		✓	62.2	65.8	0.267G	2.19M
✓	✓	✓	**64.9**	**68.2**	0.292G	2.30M

Table 6
Neck architecture comparison on AP-10K.

Neck	Fusion Type	AP	AR	FLOPs	Params
ChannelMapper	none	59.8	63.2	0.29G	3.20M
FPN [27]	top-down	58.0	61.8	0.30G	3.92M
CSPNeXt-PAFPN	multi-stage	61.5	65.0	0.66G	4.71M
YOLOX-PAFPN [28]	multi-stage	62.0	66.1	0.62G	3.64M
Ours	single-pass	**64.9**	**68.2**	**0.29G**	**2.30M**

increasing AP from 58.8 to 63.4, highlighting the importance of effective multi-scale feature calibration under lightweight constraints. In comparison, DCP and SAP provide smaller but consistent gains. DCP improves performance by enlarging the receptive field with minimal overhead, while SAP enhances feature-to-coordinate mapping for more stable predictions.

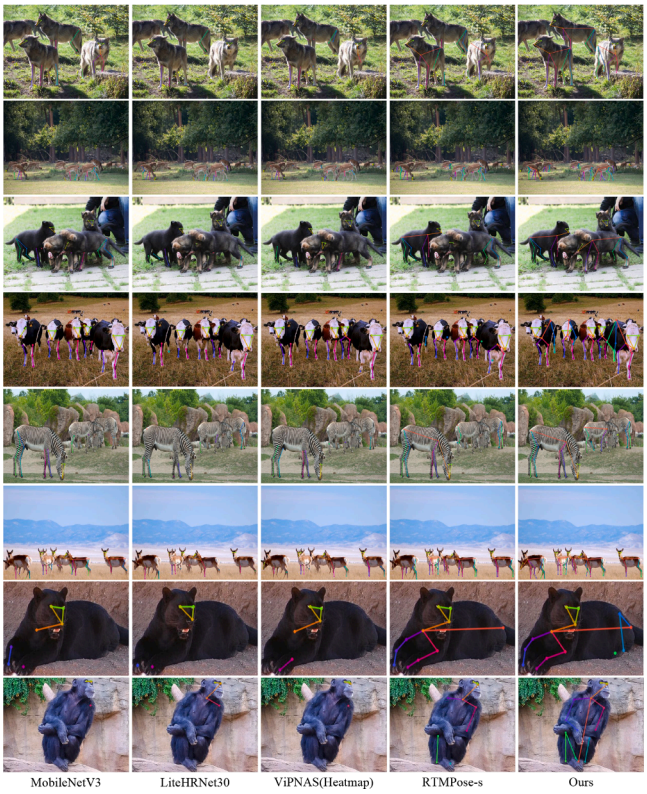


Fig. 5. AP-10K baseline comparison. Our method maintains more complete skeleton structures for single animals and identifies more instances in multi-animal scenarios.

We further compare different neck architectures, as shown in Table 6. Multi-stage fusion methods achieve reasonable performance but incur higher computational cost. In contrast, the proposed single-pass aggregation achieves higher accuracy with lower computation, indicating a more efficient multi-scale integration.

Overall, the results suggest that the proposed high-resolution calibration is the main contributor to performance gains, with additional modules providing complementary improvements.

5.3. Evaluation in real-world surveillance scenarios

We further evaluate HRCPose on CM-9k, a macaque cage-surveillance dataset, to assess its applicability in real-world scenarios. CM-9k is collected from high-resolution surveillance videos and involves complex illumination, occlusion, and pose variations, posing significant challenges for spatial detail preservation.

As shown in Table 7, HRCPose achieves the best accuracy with the lowest computational cost among lightweight models. The advantage is

Table 7
Comparisons on the CM-9k val set with input size 384 × 288.

Model	Params ↓	FLOPs ↓	AP(%) ↑	AR(%) ↑
LiteHRNet18	**1.13M**	0.60G	79.0	82.3
LiteHRNet30	1.76M	0.96G	79.8	83.7
MobileNetV2	9.57M	3.56G	80.5	83.6
ShuffleNetV2	7.55M	3.07G	79.5	82.9
RTMPose-t	3.46M	0.80G	79.3	83.6
ViPNAS(SimCC)	15.19M	1.86G	79.6	83.4
HRCPose(Ours)	2.56M	**0.49G**	**82.6**	**86.2**

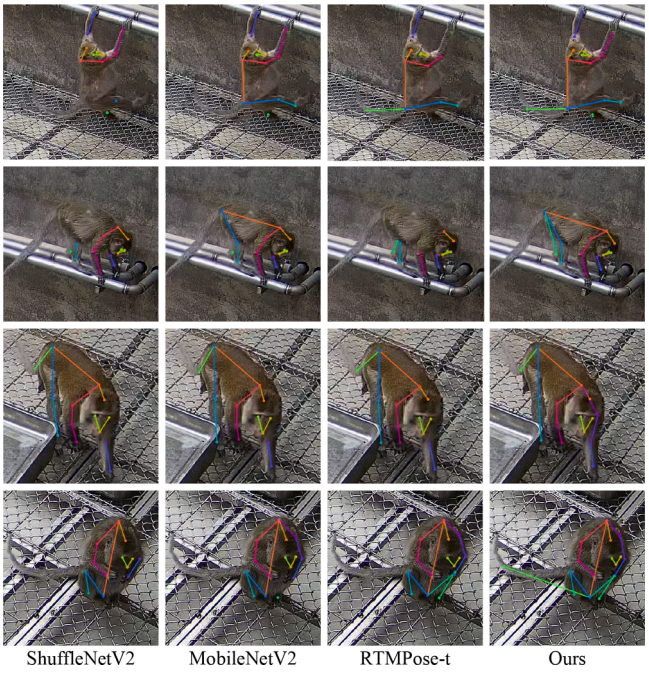


Fig. 6. CM-9k baseline comparison. Our method produces more complete and consistent skeleton predictions under challenging conditions.

more pronounced in this real-world setting, demonstrating strong generalization beyond benchmark conditions. Qualitative results in Fig. 6 further confirm this trend. Under challenging conditions, existing methods often produce missing or unstable keypoints, while HRCPose yields more complete and consistent predictions.

6. Conclusion

Experimental results on AP-10K, CM-9k, and COCO indicate that effective cross-scale interaction plays a critical role in lightweight pose estimation. By facilitating information exchange in a unified high-resolution space, the proposed framework achieves improved localization accuracy while maintaining low computational complexity. The strong performance in both benchmark and real-world surveillance scenarios suggests that the proposed design provides a practical solution for deployment-oriented animal pose estimation.

Despite its effectiveness, the current framework has two limitations: (i) it does not jointly optimize detection and pose estimation, which may introduce error propagation; (ii) it focuses on single-frame accuracy and does not explicitly model temporal consistency. Future work will explore end-to-end and temporally-aware approaches for more robust long-term monitoring.

CRediT authorship contribution statement

Jiajie Wen: Writing – original draft, Visualization, Validation, Methodology, Conceptualization; **Le Zhang:** Validation; **Haifeng Li:** Investigation; **Canbin Li:** Data curation, Investigation; **Xuejun Yue:** Writing – review & editing, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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